Year 10 Engineering Technology - Water Control System

Scope and sequence for Term 1: Garden Watering / Water Control System project

Unit Overview

Students design, construct, test and evaluate a small water-control system for a garden watering context. The unit uses the NESA Engineering Technology 7-10 Control systems focus area and draws on electronic and hydraulic control contexts. Students apply safe work practices, use input-control-output language, develop block diagrams and sketches, construct a prototype, test performance and document troubleshooting and evaluation in an engineering folio.

NESA Alignment Rule Used

- This program uses the Engineering Technology 7-10 Stage 5 outcomes coded EGT5.
- Outcome wording is taken from the local NESA Engineering Technology 7-10 syllabus document in the repository.
- The unit is aligned to the Control systems focus area. Electronic and hydraulic options are used only where they directly support the garden watering prototype.
- The program avoids unsupported outcome claims: each week includes evidence that allows the outcome to be observed, discussed or assessed.

Stage 5 Outcomes

NESA Stage 5 outcome	Official outcome wording used for this unit
EGT5-SAF-01	applies risk management and safe work practices in engineering contexts
EGT5-COM-01	communicates ideas, concepts and solutions for engineering practice
EGT5-EVL-01	investigates and evaluates engineering systems and solutions
EGT5-IVT-01	explains engineering practices and the influence of technologies used in engineering industries
EGT5-GRP-01	develops and applies technical graphics in engineering contexts
EGT5-MEA-01	applies mechanical analysis and practical testing to investigate engineering concepts
EGT5-USE-01	selects and applies materials for engineering projects
EGT5-ENV-01	analyses relationships between engineering design, production and sustainability

Control Systems Content Alignment

NESA control systems content	Where it appears in this unit
Input, control and output principles; open-loop and closed-loop systems	Weeks 2, 4 and 5 through IPO diagrams, feedback comparison and algorithm/flowchart development.
Electronic control systems: input and output components, actuators and controllers	Weeks 3, 7 and 8 through sensor/switch, controller/relay and output testing.
Hydraulic control systems: reservoir, pump, valve and actuators	Weeks 3, 6 and 8 through water path construction and pump/valve output testing.
Safe work practices in design, production and testing	Weeks 1, 2, 5, 6, 7 and 8 through wet/dry zones, teacher check points and testing routines.
Engineering process to produce a functioning control system	Weeks 4-8 through brief analysis, design, construction, integration and testing.
Testing efficiency, troubleshooting and recording solutions	Weeks 8-9 through repeated tests, response time, water delivered, fault log and evaluation.
Technical communication: sketches, diagrams, block diagrams and report	Weeks 4-5 and 9-10 through annotated sketches, block diagrams, test data and folio/report.
Sustainability and environmental considerations	Weeks 1, 4 and 9 through water efficiency, materials,

reuse/recycling and final evaluation.

10-Week Scope and Sequence

Week	Focus	Learning experiences	Target outcomes	Evidence
1	Unit launch, WHS and control-system context	Project brief; wet/dry zones; low-voltage safety; everyday irrigation and automation examples.	EGT5-SAF-01, EGT5-IVT-01, EGT5-COM-01	Safety notes, first project brief response and hazard list.
2	Inputs, controls, outputs and feedback	Input-process-output language; open-loop and closed-loop control; moisture, rain, switch and timer inputs.	EGT5-EVL-01, EGT5-COM-01, EGT5-IVT-01	IPO diagram, vocabulary notes and feedback explanation.
3	Electronic and hydraulic control components	Controllers, relays, actuators, pumps, valves, reservoirs, tubing and plant-bed layout.	EGT5-USE-01, EGT5-MEA-01, EGT5-EVL-01	Component comparison table and annotated component sketch.
4	Design brief analysis and concept development	User need; success criteria; concept sketches; water path and dry electronics zone.	EGT5-GRP-01, EGT5-COM-01, EGT5-ENV-01	Two concept sketches, selected concept and design justification.
5	Final design planning	Final system diagram; block diagram; algorithm or flowchart; component list; build sequence.	EGT5-GRP-01, EGT5-COM-01, EGT5-SAF-01	Final design plan, block diagram and risk controls.
6	Prototype construction: water path and output subsystem	Build base, reservoir, tubing and outlet; leak testing; pump/valve output testing.	EGT5-SAF-01, EGT5-USE-01, EGT5-MEA-01	Build log, output test data and teacher check record.
7	Prototype construction: input and control subsystem	Sensor/switch testing; relay/controller setup; wiring checks; threshold or trigger testing.	EGT5-SAF-01, EGT5-EVL-01, EGT5-MEA-01	Input test data, wiring photo/diagram and troubleshooting notes.
8	Whole-system testing and troubleshooting	Connect subsystems; run repeated tests; record response time, water delivered, faults and fixes.	EGT5-EVL-01, EGT5-MEA-01, EGT5-COM-01	Full-system test table and fault log.
9	Evaluation, sustainability and improvement	Reliability, efficiency, water use, materials, repairability, recycling/reuse and redesign decisions.	EGT5-ENV-01, EGT5-EVL-01, EGT5-COM-01	Evaluation draft using test evidence.
10	Folio completion and project presentation	Engineering report structure; peer review; final reflection; submission preparation.	EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01	Completed folio/report and short project presentation.

Assessment and Evidence

- Prototype: a functioning water-control system that demonstrates input, control/process and output.
- Folio/report: brief analysis, research, IPO/block diagram, sketches, final design, algorithm, risk controls, build log, test data, troubleshooting and evaluation.
- Teacher observation: safe work practices, tool/component use, testing behaviour and collaboration.

- Evaluation: reliability, efficiency, safety, sustainability, water use and justified improvements.

Differentiation and Adjustments

- Support: provide partially completed diagrams, vocabulary banks, step-by-step build cards and paired testing roles.
- Core: students independently produce diagrams, testing records and evidence-based evaluation.
- Extension: add closed-loop feedback, compare alternative sensors/outputs, improve water efficiency or graph repeated test data.
- Adjustments should support access to Stage 5 outcomes before considering Life Skills outcomes, consistent with NESA guidance.

Required Resources

- Low-voltage power only, teacher-approved controller/relay/circuit option, sensors or switches, pump or solenoid valve, reservoir, tubing, base board, plant-bed tray, measuring cylinder, timer, towels/spill kit and PPE.
- Website pages: Year 10 Term 1 landing page, toolkits 1-4, tutorial, testing checklist, info sheets and portfolio workbook.
- Local source: NESA Engineering Technology 7-10 syllabus, Control systems focus area.